FIRST-MOVE ADVANTAGE IN ONLINE CHESS ACROSS RATING BRACKETS

AUTHORS

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GITHUB

https://github.com/TrtQu/First-Move-Advantage-in-Online-Chess.git

Does White's edge from moving first change with player skill and time control? Answered with all 86.5 million rated Lichess games from June 2026.

INTRODUCTION AND MOTIVATION

Chess players have known for over a century that White enjoys a first-move advantage, but does that edge hold at every skill level, or is it something only strong players take advangage of? And does it change with time control? Excel and pandas both break well before you can answer that question at scale, so we profiled, broke, and then scaled past the full Lichess rated games dump for June 2026 with command-line tools and PySpark on Google Cloud Dataproc.

26 GB

compressed size (pgn.zst)

1.73B

lines of data after decompression

86.48M

full rated chess games analyzed

PROFILING

Dataset Overview:

- Size: ~148 GB uncompressed
- Rows: 1.7 billion (from wc -l count)

Key Data Insights:

- Average White Elo: 1643.37 computed in 14m 24s
- Top opening: Queen's Pawn Game

Cost of Sequential Streaming:

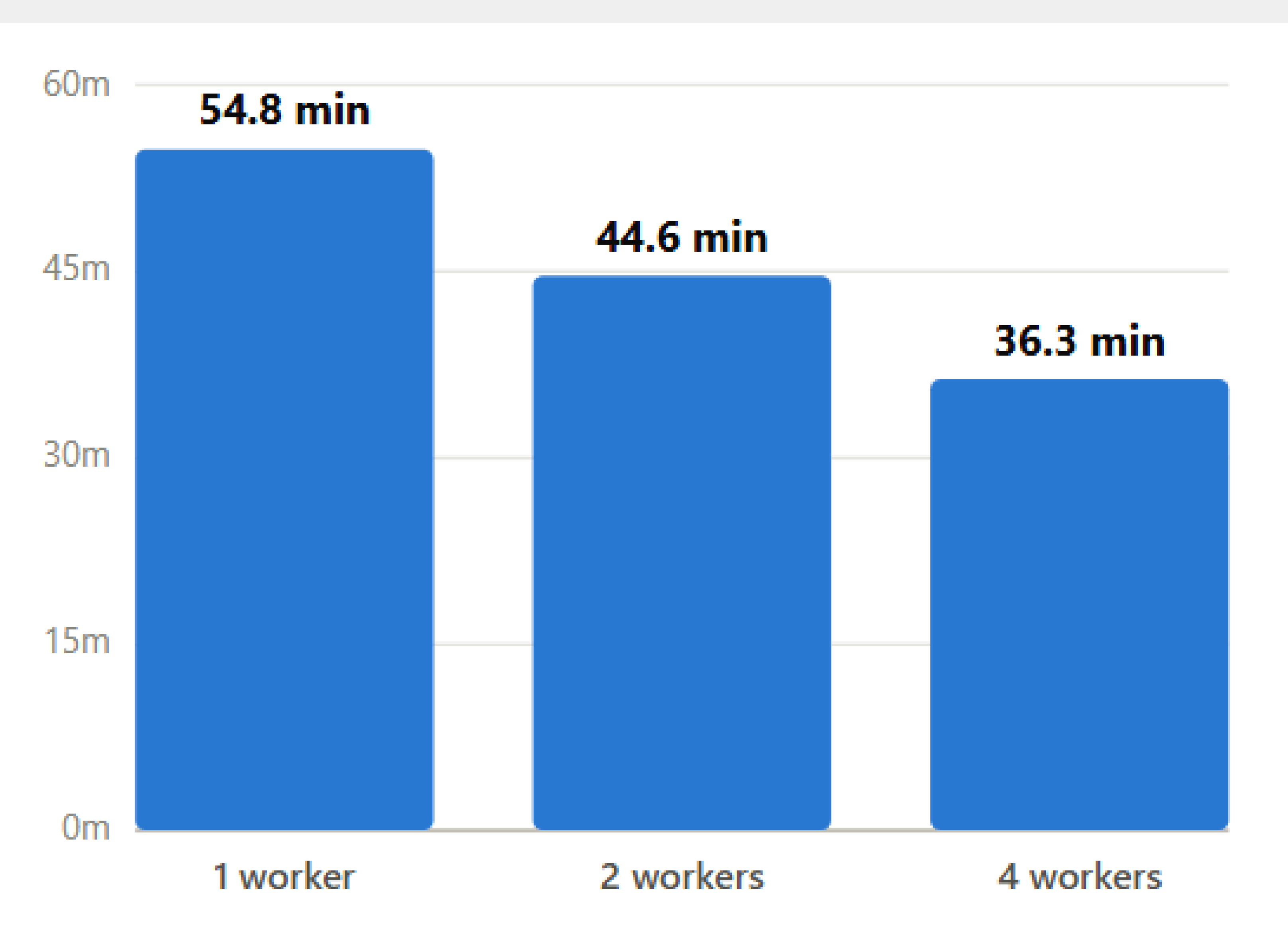
- Reading the start: 0.03 seconds
- Reading the end: 59.7 minutes
- Sorting openings: 22.2 minutes

Characterized the raw dataset with CLI tools only: du, wc -l, head/tail, cut, sort | uniq -c, awk, grep -c — every command streamed through zstdcat and timed with time

We never used RAM here.

SCALING

The Same PySpark Job (explicit schema, transforms, groupBy, a window function, a join function, caching, and a Spark SQL cross-check) reads the full dataset directly from GCS and runs unchanged on Dataproc clusters of 1, 2, and 4 workers

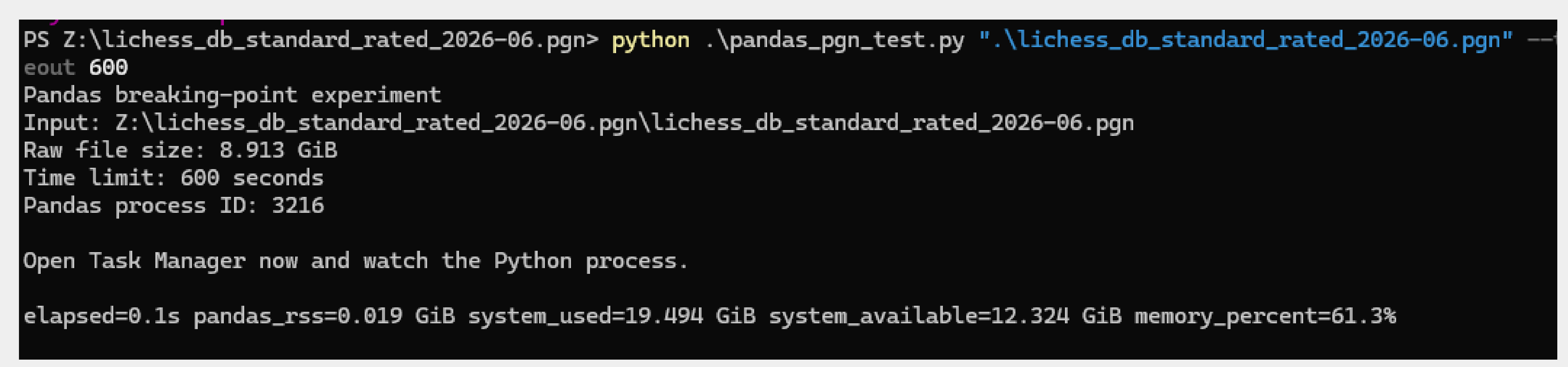


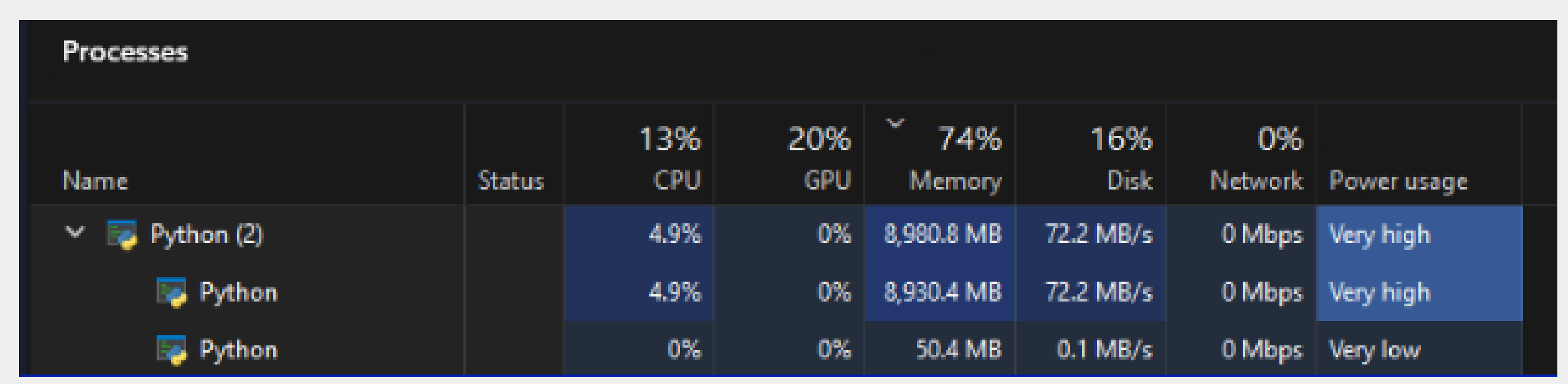
runtime only falls ~18.6% per doubling, not ~50% linear scaling would predict

BREAKING

Pandas Test

read_csv() with no chunksize on the ~8.9 GB decompressed PGN pulled the entire file into RAM before doing anything else. Peak RSS 8.80 GiB, then crashed on a bad byte after 346s

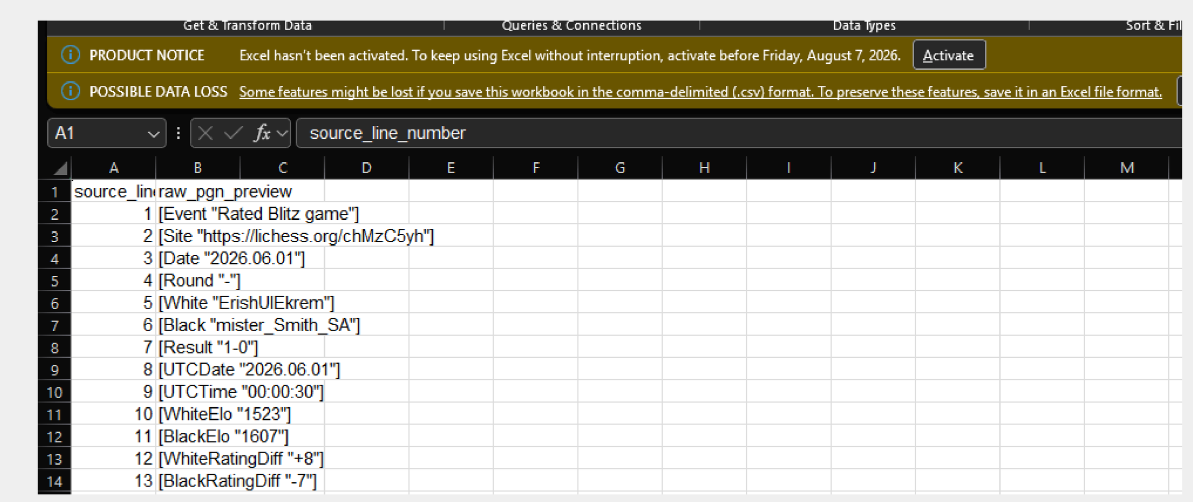


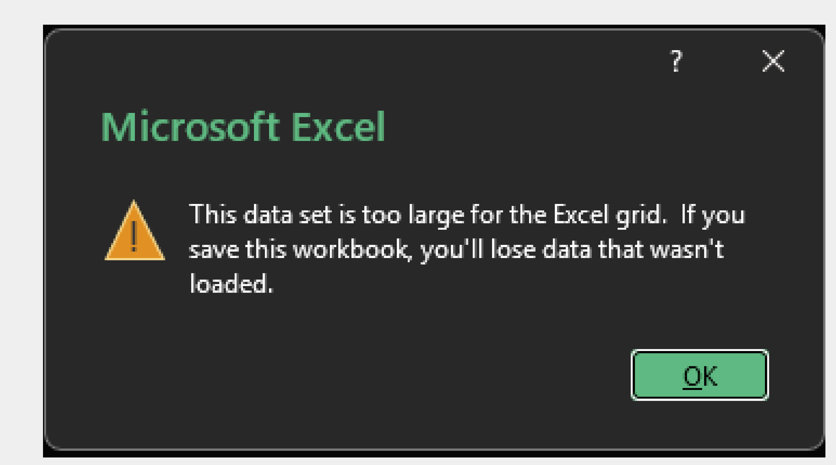


CLI vs. Spark cross-check: grep -c counted 86,483,328 games in 14m24s; the Spark job independently landed on 86,458,368. This is a 0.03% gap explained by Spark's stricter row filter.

Excel Test

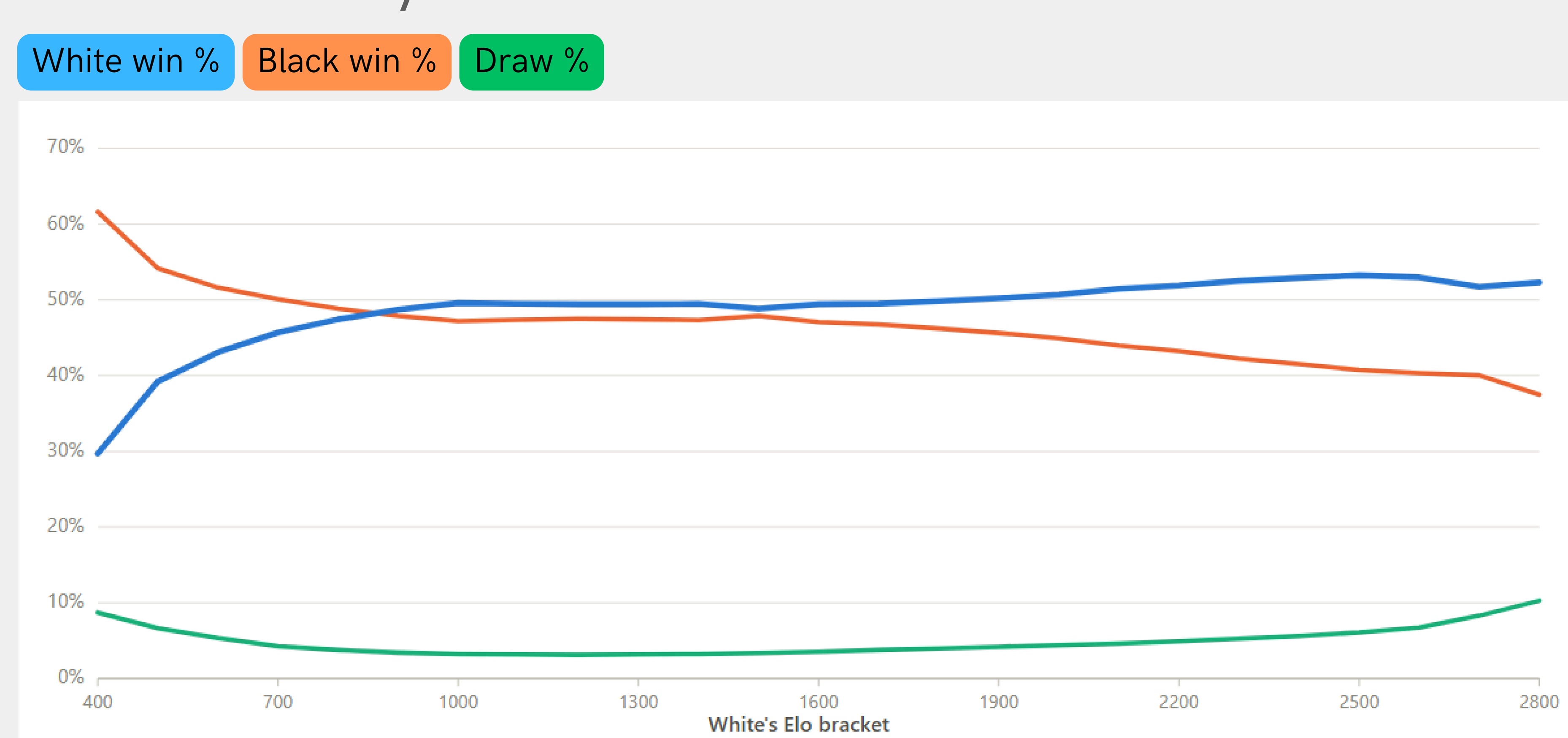
A 400 MB/ 10,050,001 row CSV silently truncates to Excel's 1,048,576 row cap, then throws a warning





"This data set is too large for the Excel grid"

RESULTS/FINDINGS



Skill matters. White's overall edge barely moves but climbs steadily with player strength. Weak players convert the first-move advantage only weakly. At the lower brackets, Black actually wins more, likely from White blundering the extra tempo away.

29.7% → 52.9%

White's win rate at the lowest Elo bracket(400 vs. master level (2400+)

61.6% → 41.5%

Black's win rate falls by the mirror amount, crossing White's line at around 900-1000 Elo

Time control	White win %	Draw %	Games
Bullet	49.99%	2.87%	31.7M
Blitz	49.59%	4.32%	40.1M
Rapid	49.54%	4.66%	14.2M
Classical	48.82%	5.99%	383K